

Propositions, $((x \in A \wedge y \in B) \wedge x \in A) \wedge (y \in C)$. By 2nd Commutative Law of Propositions, $(x \in A \wedge (x \in A \wedge y \in B)) \wedge (y \in C)$. By 2nd Associative Law of Propositions, $((x \in A \wedge x \in A) \wedge y \in B) \wedge y \in C$. By 2nd ~~Commutative Law of~~ Idempotent Law of Propositions, $(x \in A \wedge y \in B) \wedge y \in C$. By 2nd Associative Law of Propositions, $(x \in A \wedge (y \in B \wedge y \in C))$. By Defn. 2, $x \in A \wedge (y \in B \cap C)$. By Defn. 8 from Sec-2.1, $(x, y) \in A \times (B \cap C)$. By Defn. 3 from Sec-2.1, $(A \times B) \cap (A \times C) \subseteq A \times (B \cap C)$

$$\therefore A \times (B \cap C) = (A \times B) \cap (A \times C) \quad \square$$

37) Prove or disprove that for all sets A, B and C, we have

$$a) A \times (B - C) = (A \times B) - (A \times C) \quad (\text{Defn. 8 from Sec-2.1})$$

Solution:

$$\begin{aligned}
 A \times (B - C) &= \{(x, y) \mid x \in A \wedge y \in (B - C)\} \quad (\text{Defn. 4}) \\
 &= \{(x, y) \mid x \in A \wedge (y \in B \wedge y \notin C)\} \quad (\text{Defn. 4}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge y \notin C\} \quad (\text{2nd Associative Law of Proposition}) \\
 &= \{(x, y) \mid (x \in A \wedge x \in A) \wedge y \in B \wedge y \notin C\} \quad (\text{2nd Idempotent Law of Proposition}) \\
 &= \{(x, y) \mid (x \in A \wedge (x \in A \wedge y \in B)) \wedge y \notin C\} \quad (\text{2nd Associative Law of Proposition}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge x \in A \wedge y \notin C\} \quad (\text{2nd Commutative Law of Proposition}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge (x \in A \wedge y \notin C)\} \quad (\text{2nd Associative Law of Proposition}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge (x \in A \wedge \neg(y \in C))\} \quad (\text{Defn. 5}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge (x \in A \wedge \neg(y \in C))\} \quad (\text{Defn. 8 from Sec-2.1}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge (x \in A \wedge \neg(y \in C))\} \quad (\text{By Defn. of not belongs to}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B) \wedge (x \in A \wedge \neg(y \in C))\} \quad (\text{By Defn. of AND}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B), x \in A, \neg(y \in C)\} \quad (\text{By Defn. of AND}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B), x \in A, \neg(y \in C) \text{ is False}\} \quad (\text{Defn. of negation}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B), x \in A, y \in C \text{ is False}\} \quad (\text{Double-negation Law of Propositions}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B), (x \in A \wedge y \in C) \text{ is False}\} \quad (\text{Defn. of AND}) \\
 &= \{(x, y) \mid (x \in A \wedge y \in B), (x \in A \times C) \text{ is False}\} \quad (\text{Defn. 8 from Sec-2.1})
 \end{aligned}$$